

integration.

Transformers, FAISS, Streamlit, SQLAlchemy, and Google Gemini API for LLM

Technical implementation uses Python with Pandas, NumPy, Sentence quality across completeness, consistency, validity, and accuracy dimensions. and invalid values with severity-based scoring. The system evaluates data

Quality assessment includes detecting missing values, duplicates, outliers, consistent representation.

specific loaders for each data type and a common normalization layer for data formats including CSV, Excel, JSON, XML, TXT, and PDF. It uses format-

The system follows a modular pipeline architecture that supports multiple and actionable correction recommendations.

(RAG) to provide intelligent, context-aware explanations of data quality issues traditional data profiling techniques with modern retrieval-augmented generation assessment system designed for enterprise data lakes. The system combines

This document provides a comprehensive overview of the data quality Data Quality Assessment System for Enterprise Data Lakes